

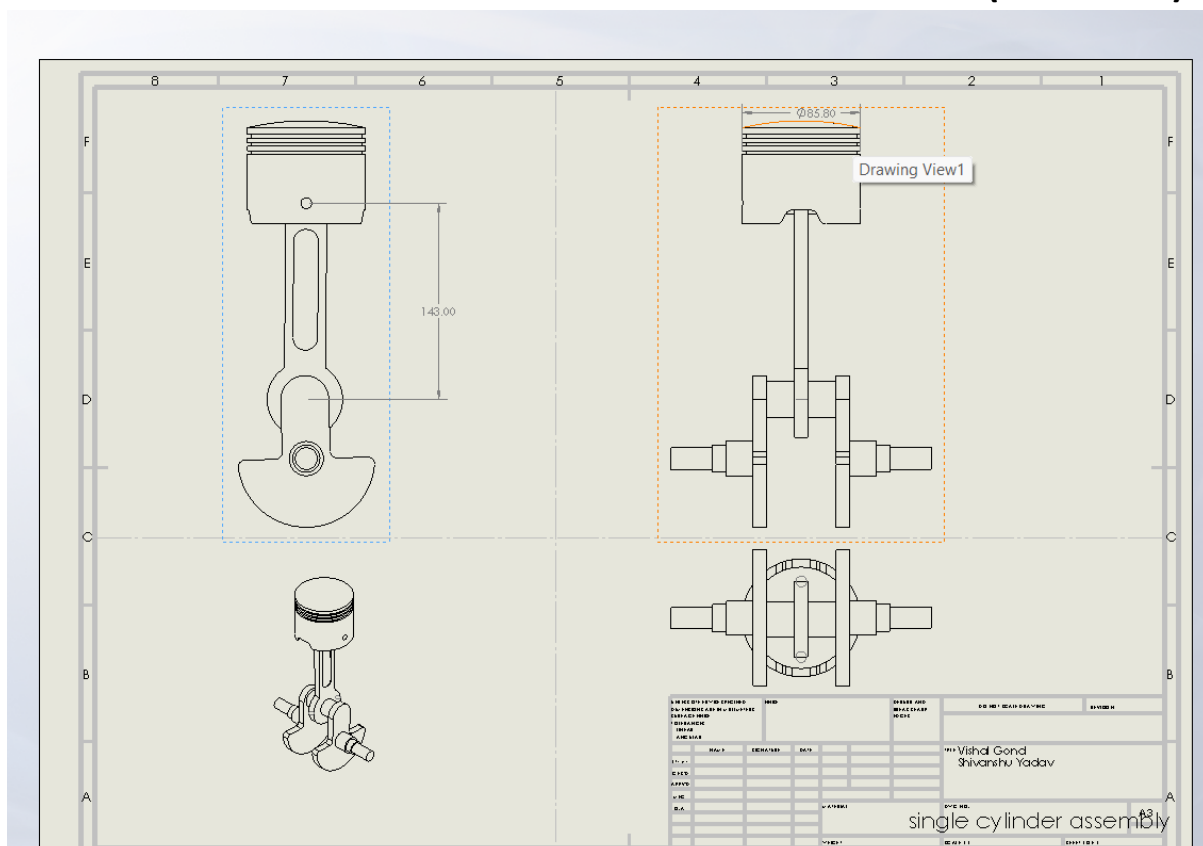
Analysis of Single-cylinder engine assembly

ANSYS Workbench Presentation

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Bore = 86 mm

Stroke = 86 mm

Connecting Rod Length = 143 m

Pressure = 6 MPa

Materials:

Piston : Aluminium Alloy

Connecting Rod: Steel 4340

Block: Aluminium Alloy

MODAL ANALYSIS

Modal analysis is used to determine the **natural frequencies** and **mode shapes** of the engine components.

It tells how the piston, connecting rod, crankshaft, and pins vibrate when excited by dynamic forces.

A piston–rod–crank mechanism is subjected to:

Combustion forces, Road-induced vibrations, Crankshaft imbalance, Structural engine noise

These create dynamic loads.

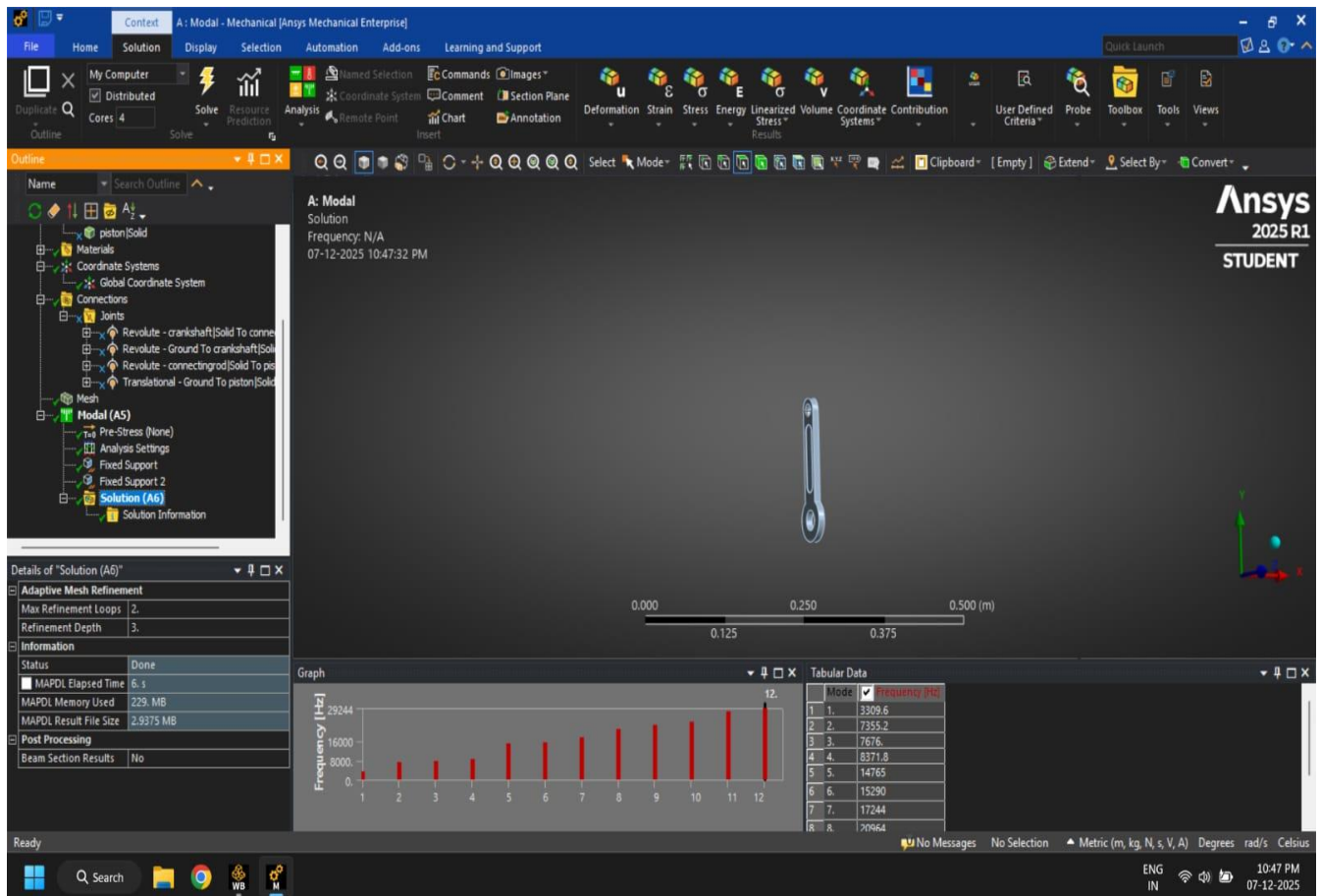
If these loads match the natural frequency → **Resonance occurs.**

Resonance causes:

Excessive vibration, Noise, Fatigue failure, Crankshaft bending, Rod or pin fracture

So modal analysis is critical for engine durability.

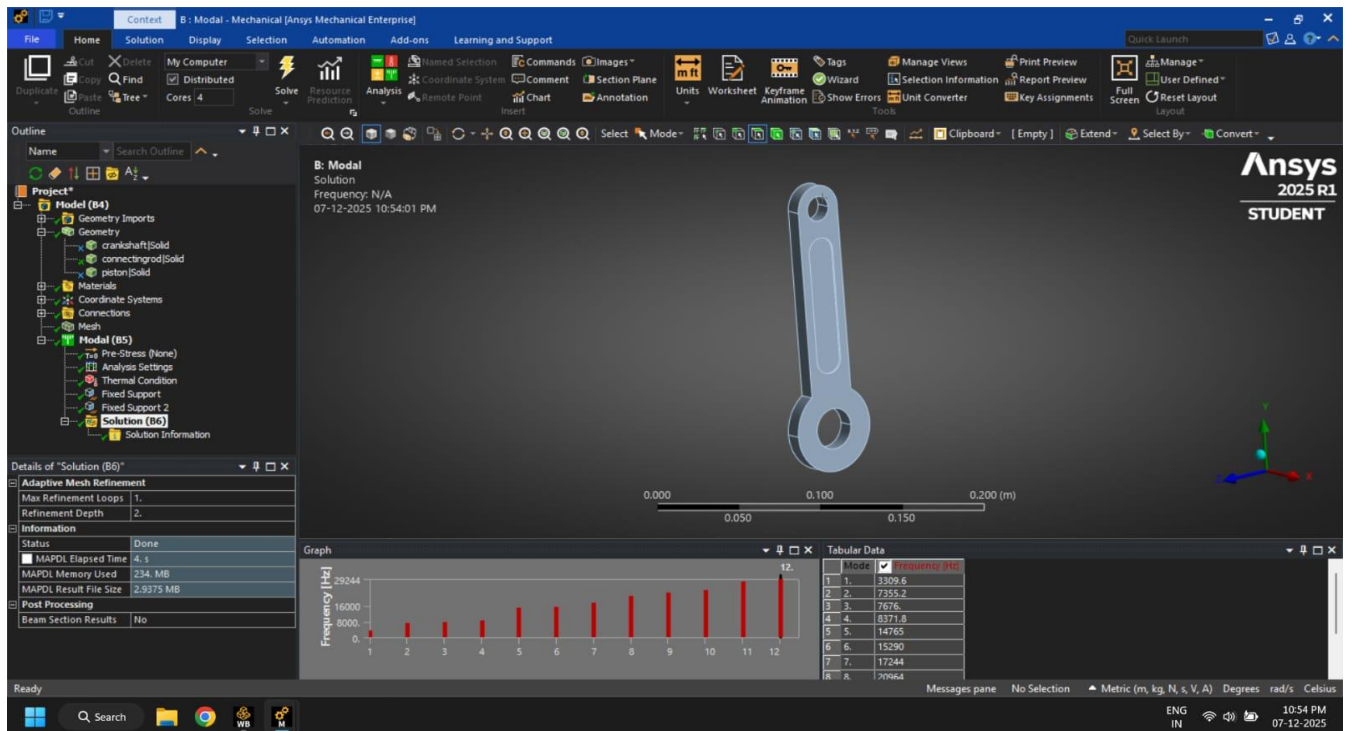
Case A:- Cold Condition :- No thermal preload



Tabular Data above represent different bending and twisting vibration shapes of the connecting rod and assembly.

3309 Hz indicates the lowest vibration frequency of the connecting rod which is well **above engine operating frequency** (16–100 Hz), so **no resonance risk** during normal engine RPM (1000-6000 rpm).

Case B:- Hot Condition



Under hot operating temperature, the connecting rod shows slightly reduced stiffness, but all natural frequencies remain **very high (3300–21000 Hz)**. This means that even after thermal expansion and loss of material stiffness, the structure stays rigid and stable. All frequencies are far above the engine excitation range (16–100 Hz), so **no resonance risk** exists during real engine operation.

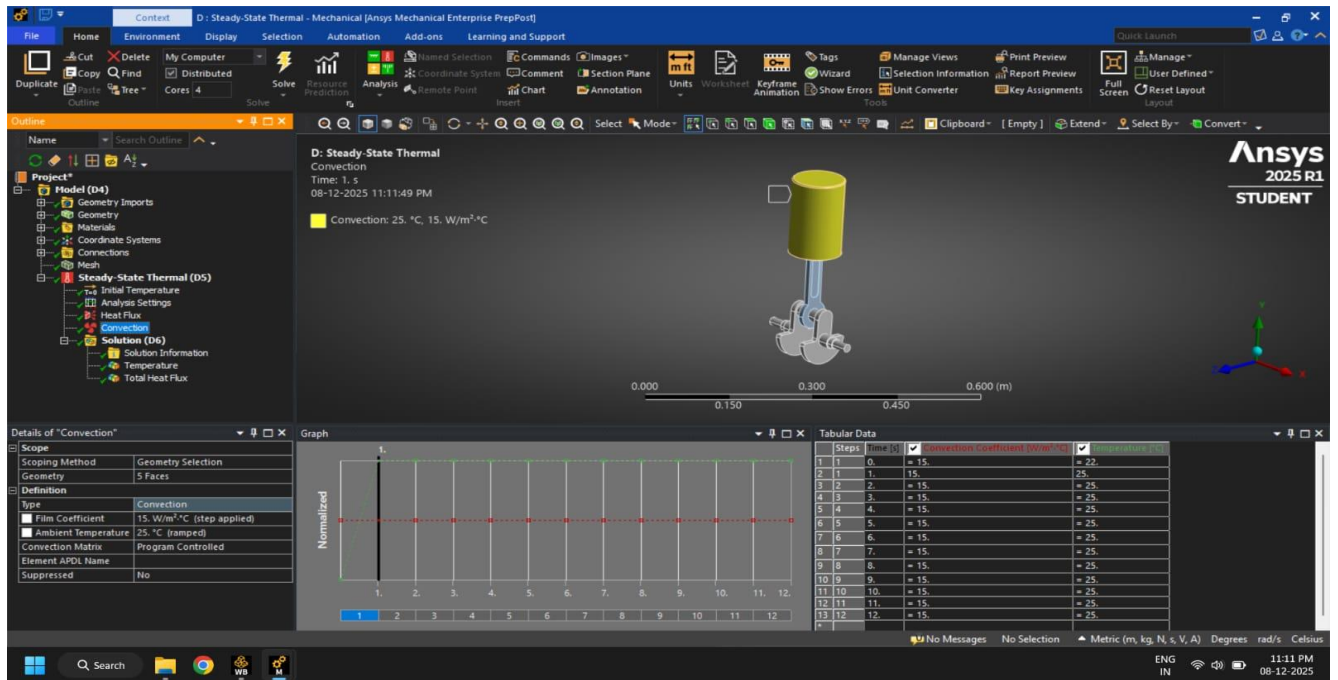
STEADY-STATE THERMAL ANALYSIS

Steady-state thermal analysis is used to find the **constant temperature distribution** and **heat flow** in a body after **thermal equilibrium** is reached.

It is needed to know **final operating temperature** of components, check **overheating**, **thermal stresses**, and material safety and ensure proper **cooling**, **insulation**, and **heat dissipation** design.

Steady-state thermal analysis determines the **constant temperature field** under given thermal loads to ensure the **safe thermal performance** of components.

Case 1 Outer Block Convection



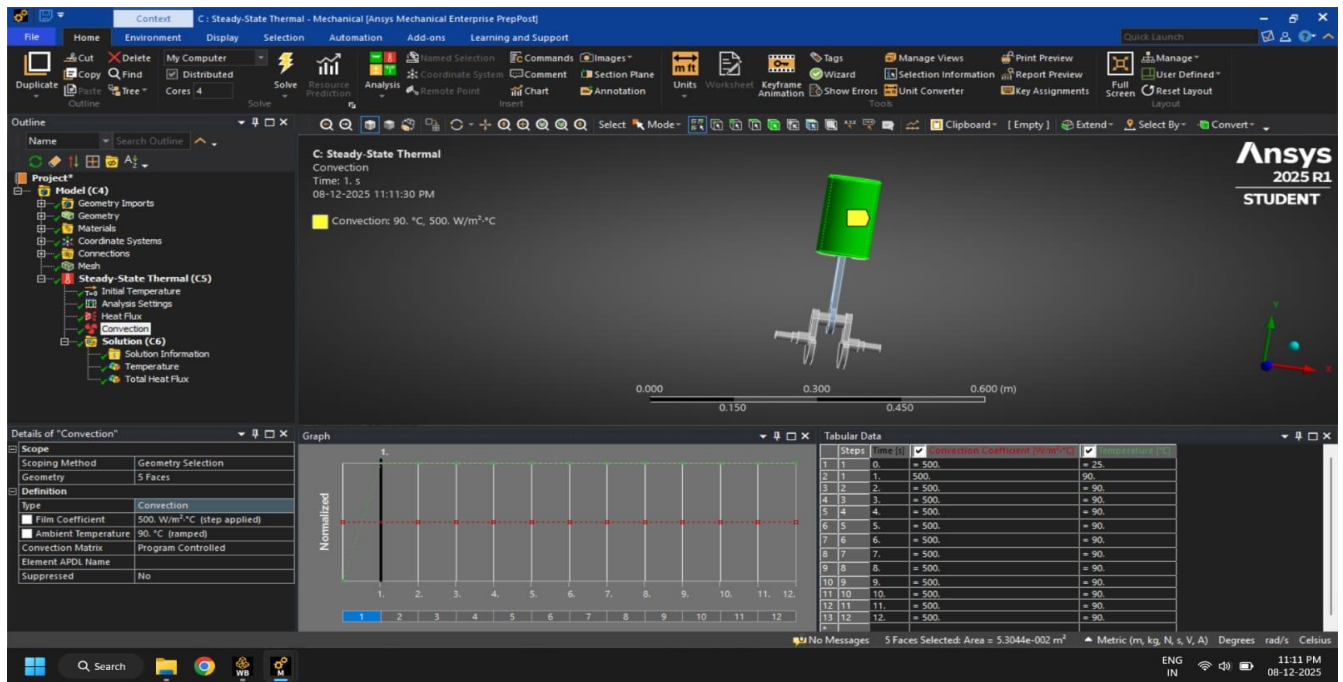
Higher heat flux → higher surface temperature

Low convection (15 W/m²·°C is very low) → temperatures will rise significantly

Maximum temperature will be where heat is applied

Minimum temperature near convection surfaces

Case 2: Coolant Jacket

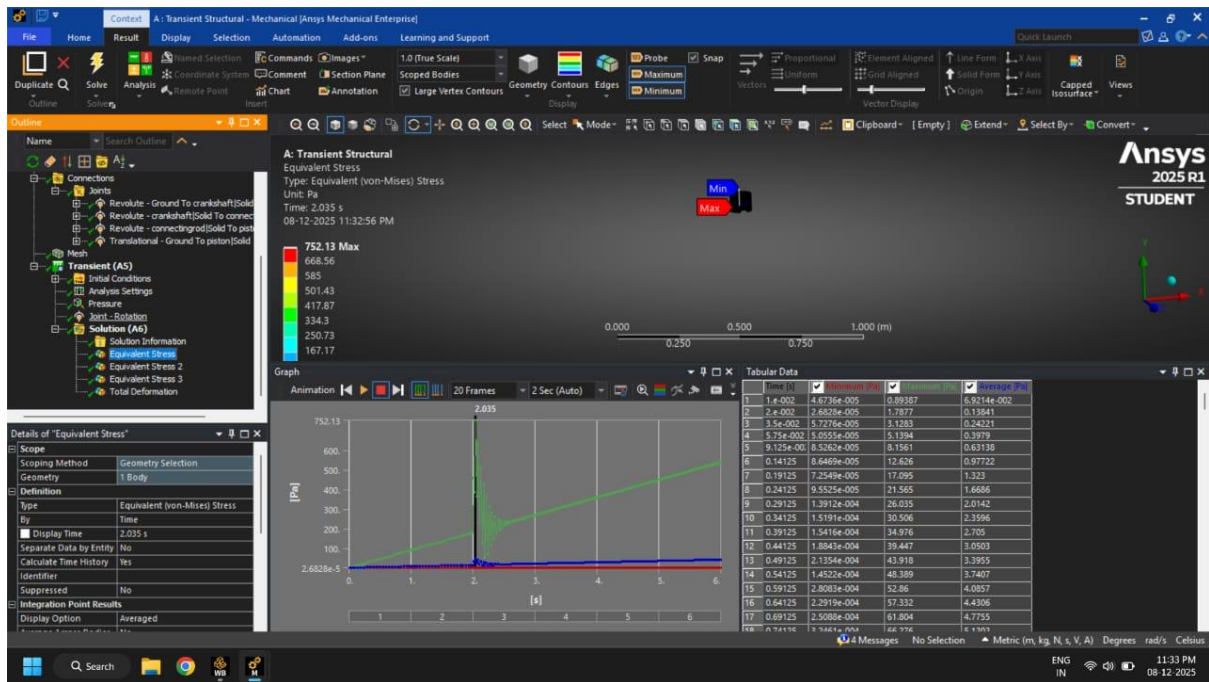


TRANSIENT STRUCTURAL ANALYSIS

A Transient Structural Analysis calculates **how a structure deforms and stresses change over time** when subjected to time-dependent loads.

We use Transient Structural Analysis because real engineering systems experience **time-dependent loads**, and only transient analysis can capture: Inertia effects, Damping, behavior, Vibrations and oscillations, Impact forces and shock waves, Peak stress response, Dynamic stability

Static analysis gives a single snapshot; transient analysis gives the **true physical behaviour** of the system.

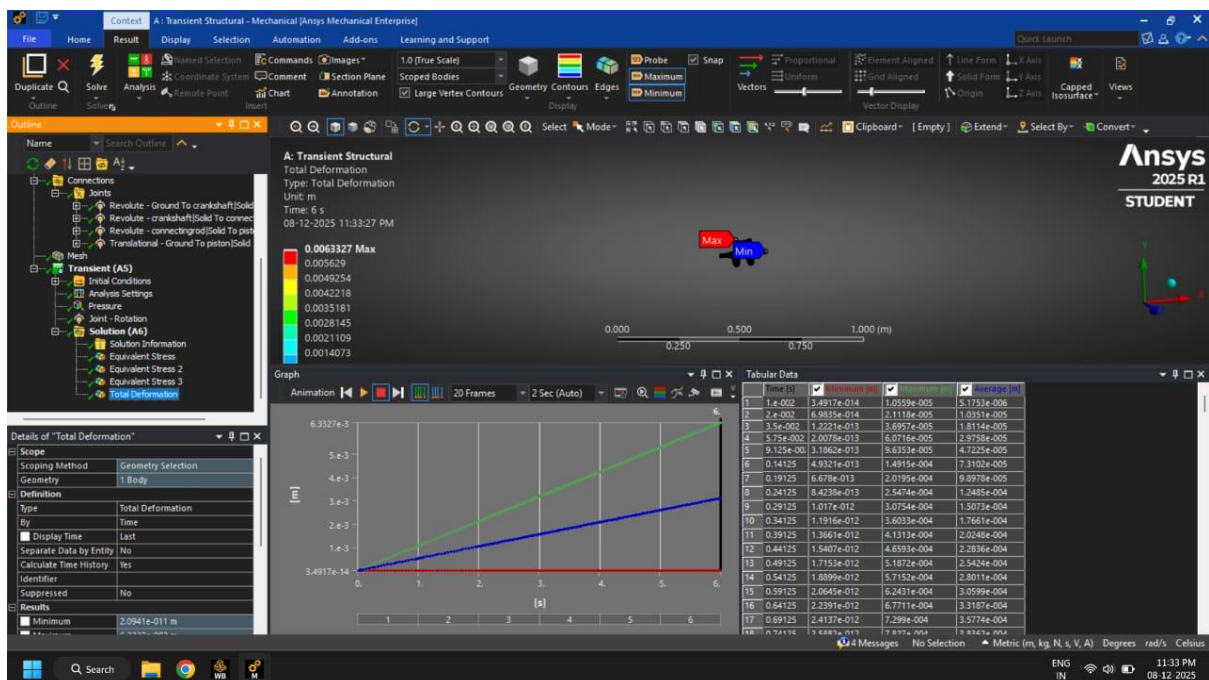


Plot of **Equivalent (von-Mises) Stress** versus **Time** in a *Transient Structural* analysis of piston.

Maximum stress ≈ 752.13 Pa

Minimum stress $\approx \sim 0$ Pa

The peak occurs at **time = 2.035 s**



Plot of **Total Deformation** over the entire transient analysis of crankshaft.

Maximum deformation: 0.0063327 m (≈ 6.3 mm)

Minimum deformation: $\sim 3.49 \times 10^{-13}$ m (basically zero)